Similarity

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Recall and choose your starting point

> Record your starting explanation for Similarity before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Confidence Is Not Correctness

Without opening the old lesson, explain one key idea from Confidence Is Not Correctness. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Boundaries and 'Not Sure'

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Learning from Examples

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Similarity trace and record the first step you can explain without help.

Use W2: Explain a changed winner for **Similarity** and selected features. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Similarity explanation against the supplied facts before discussing it.

Use the worked case for **Similarity** and selected features. Proposed explanation: “Similar has one objective meaning for every task”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Transfer the comparison to books

A target book has topic space, format comic and language English. **Candidate**s are J: space, prose, English; K: animals, comic, English; L: space, comic, Spanish. Give 1 point per exact match and return all tied highest scorers.

Then make topic worth 3 points while keeping the other **weight**s at 1. Report both sets of totals and winners. Explain why a winner may still fail a requirement to read English.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Similarity target to a trace and a justified conclusion.

A reader wants mystery and short books. A has both traits, B only mystery, C only short. Score one point per matching trait. The reader now says short is essential. Explain how the decision should change.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Similarity case and show the deciding evidence.

Book A has animals and adventure; B has animals only; C has neither. A reader wants both traits.

Rank by number of matching traits.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Compare books using topic only. Book A is about space with a red cover; B is about space with a blue cover; C is about dogs with a red cover. Which matches A best?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Similarity adds to Helpful Classroom Sorter and keep the evidence.

Choose two useful item traits for **similarity**-based sorting and explain why each deserves its **weight**.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A reader wants mystery and short books. A has both traits, B only mystery, C only short. Score one point per matching trait. The reader now says short is essential. Explain how the decision should change.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Show the contributions before the total

> Show the contributions before the total. Apply the supplied Similarity facts and keep your reasoning.

For the red round smooth target and **candidate**s A, B and C in the lesson, record the colour, shape and texture contributions under one point per match. Give each total and identify every highest scoring **candidate**.

My reasoning and evidence

W2 Explain a changed winner

> Explain a changed winner. Apply the supplied Similarity facts and keep your reasoning.

Keep the same target and **candidate**s but make texture worth 3 points, with colour and shape worth 1 each. Recalculate A, B and C. Explain what changed and what stayed the same.

My reasoning and evidence

W3 Keep unknown separate from different

> Keep unknown separate from different. Apply the supplied Similarity facts and keep your reasoning.

Candidate E is red and round but its texture is missing. A learner assigns zero for texture and declares E fully scored. Apply our classroom policy.

State the known contributions, the missing information and what must happen before a final ranking.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Transfer the comparison to books

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer the comparison to books. Apply the supplied Similarity facts and keep your reasoning.

A target book has topic space, format comic and language English. **Candidate**s are J: space, prose, English; K: animals, comic, English; L: space, comic, Spanish. Give 1 point per exact match and return all tied highest scorers.

Then make topic worth 3 points while keeping the other **weight**s at 1. Report both sets of totals and winners. Explain why a winner may still fail a requirement to read English.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Notice an accidental extra weight

> Notice an accidental extra weight. Apply the supplied Similarity facts and keep your reasoning.

A designer adds a second colour column that copies the first and gives both columns one point. They say no feature has been given extra importance because every column has equal **weight**.

Explain the problem and propose a clearer rule.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Similarity

> A fictional craft-card matcher compares each candidate with a target. Explain the labelled

> step before checking the changed condition.

A fictional craft-card matcher compares each **candidate** with a target.

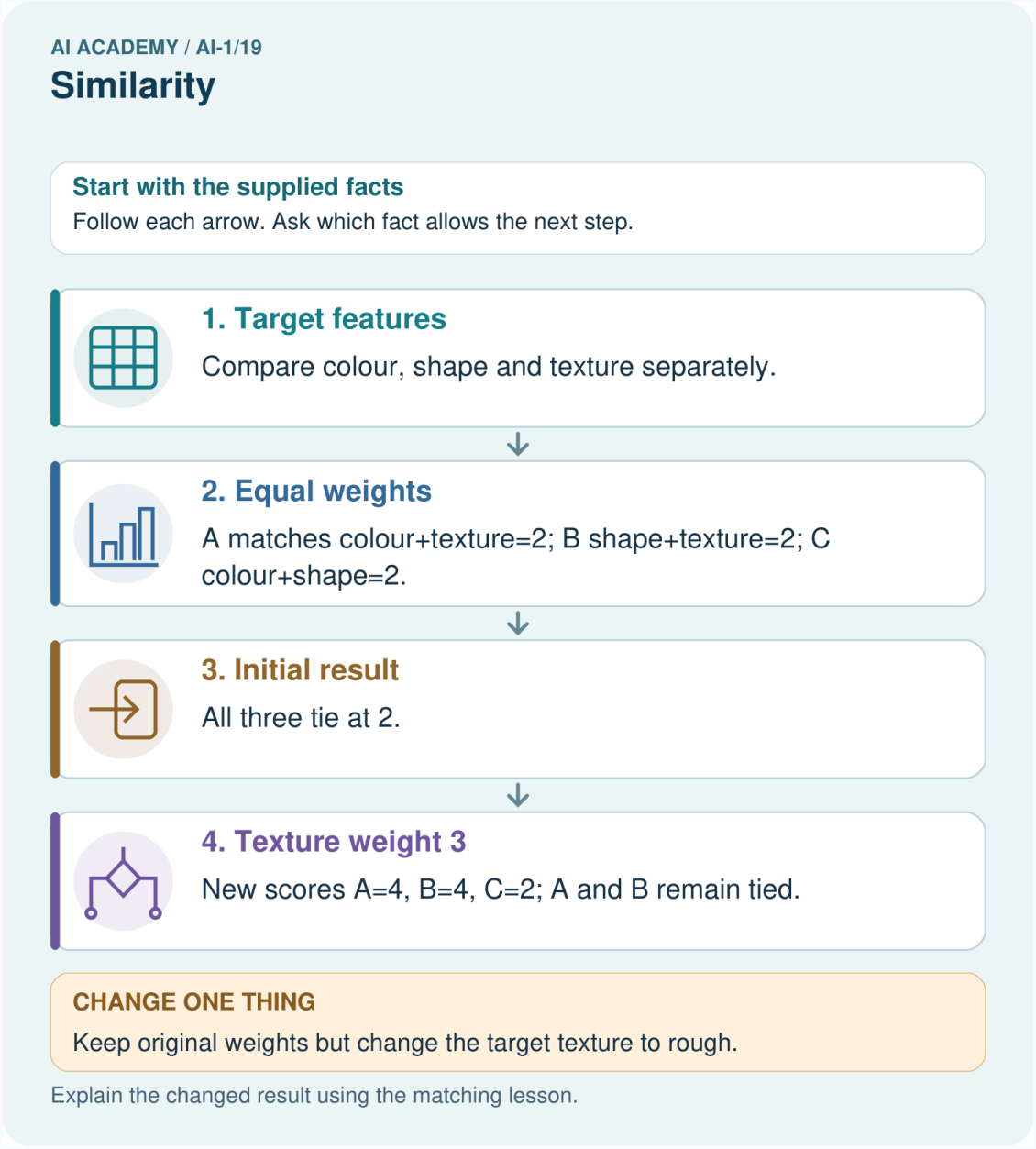
Target: red, round, smooth. A: red,square,smooth. B: blue,round,smooth. C: red,round,rough.

Initially each exact **feature match** scores 1; return all highest scorers.

For the second comparison only, colour and shape matches score 1 each and a texture match scores 3.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Point to the matches supporting C's new score and explain why there is no tie.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Declare sorter **similarity** features, **weight**s and tie behavior before using scores to explain recommendations.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Declare sorter **similarity** features, **weight**s and tie behavior before using scores to explain recommendations.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional craft-card matcher compares each **candidate** with a target.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? New scores A=4, B=4, C=2; A and B remain tied.

1. Target features

2. Initial result

3. Texture **weight** 3

4. Equal **weight**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Texture **weight** 3 New scores A=4, B=4, C=2; A and B remain tied. **Similarity** is defined by the chosen representation and **weight**s.

The cards did not change when texture gained **weight**; the comparison rule did. Two high scores need not mean two identical cards.

A complete rule also says what to do with tied scores, rather than choosing whichever appears first by accident.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Keep original **weight**s but change the target texture to rough. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. C scores 3; A and B each score 1.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: C scores 3; A and B each score 1. C now matches all three target features; the others lose their smooth-texture match.

Repair a missing building block

Review Confidence Is Not Correctness (ai-1/18). Goal: Distinguish a confidence claim from **evidence** that a **prediction** was correct. Short practice: Was the unknown label guaranteed correct because its confidence was **80%**?.

Return to this week and explain what you changed.

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a **prediction**. Short practice: Which original block needs a better width before accepting or rejecting it?

What should the reviewer do?. Return to this week and explain what you changed.

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing **prediction**s and counting disagreements for every **candidate**. Short practice: Which new measured length would reveal disagreement between the **candidate**s?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Similarity picture from its labels, then record a relevant limit.

01. Similarity: the case

AI ACADEMY / AI-1/19 · 01 · LOOK AT THE FACTS

Similarity: the case

The situation

A fictional craft-card matcher compares each candidate with a target.

Supplied evidence

Target: red, round, smooth. A: red,square,smooth. B: blue,round,smooth. C: red,round,rough.
Initially each exact feature match scores 1; return all highest scorers.
For the second comparison only, colour and shape matches score 1 each and a texture match scores 3.

By the end, I can calculate similarity under a declared feature rule and explain a changed ranking.

Follow the labelled picture with the matching explanation. Open the vector diagram

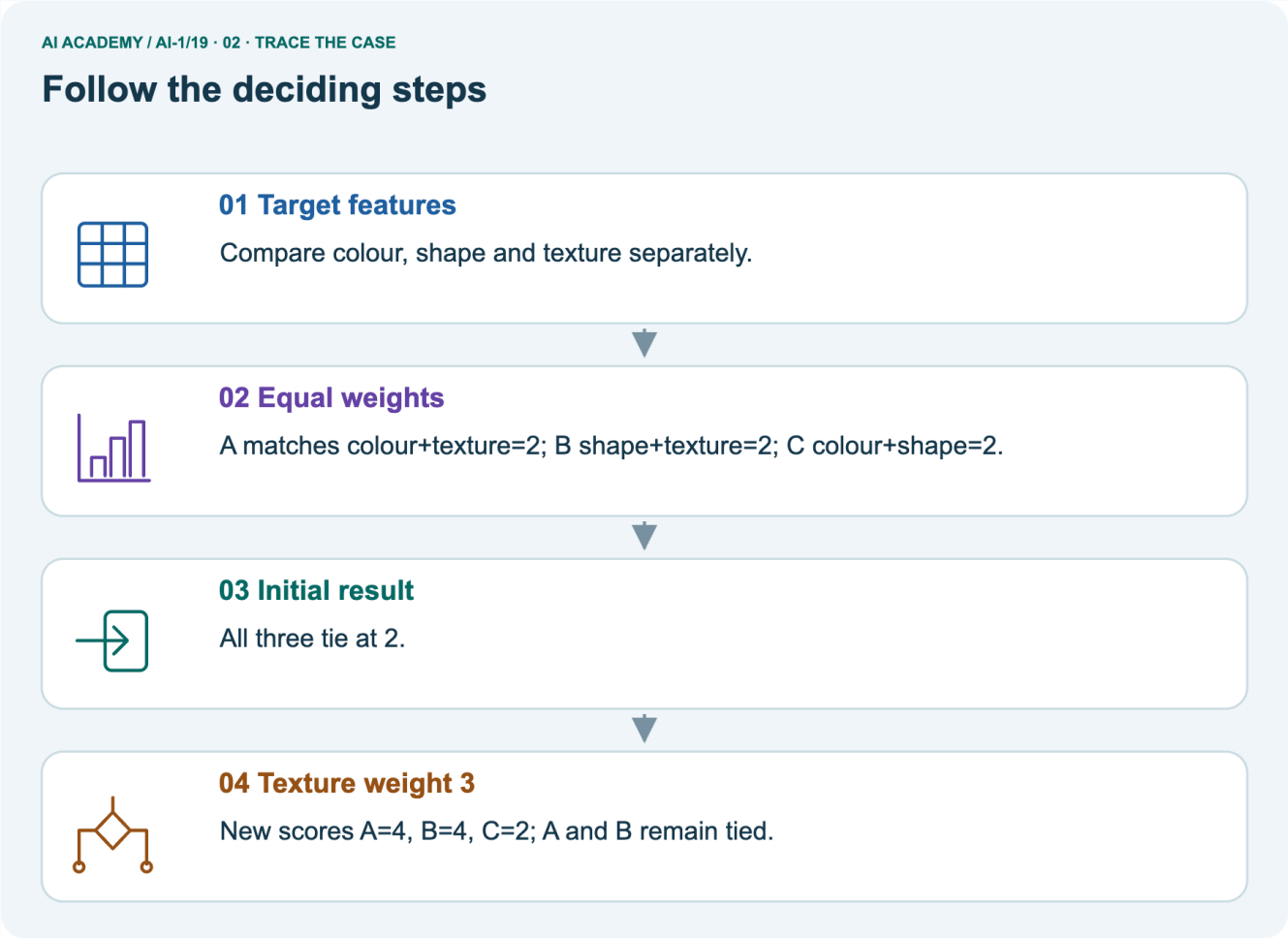
A fictional craft-card matcher compares each **candidate** with a target. Supplied facts: Target: red, round, smooth. A: red,square,smooth. B: blue,round,smooth. C: red,round,rough.

Initially each exact **feature match** scores 1; return all highest scorers. For the second comparison only, colour and shape matches score 1 each and a texture match scores 3.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Target features: Compare colour, shape and texture separately. → Equal **weight**s: A matches colour+texture=2; B shape+texture=2; C colour+shape=2. → Initial result: All three tie at 2. → Texture **weight** 3: New scores A=4, B=4, C=2; A and B remain tied.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Declared weights change similarity



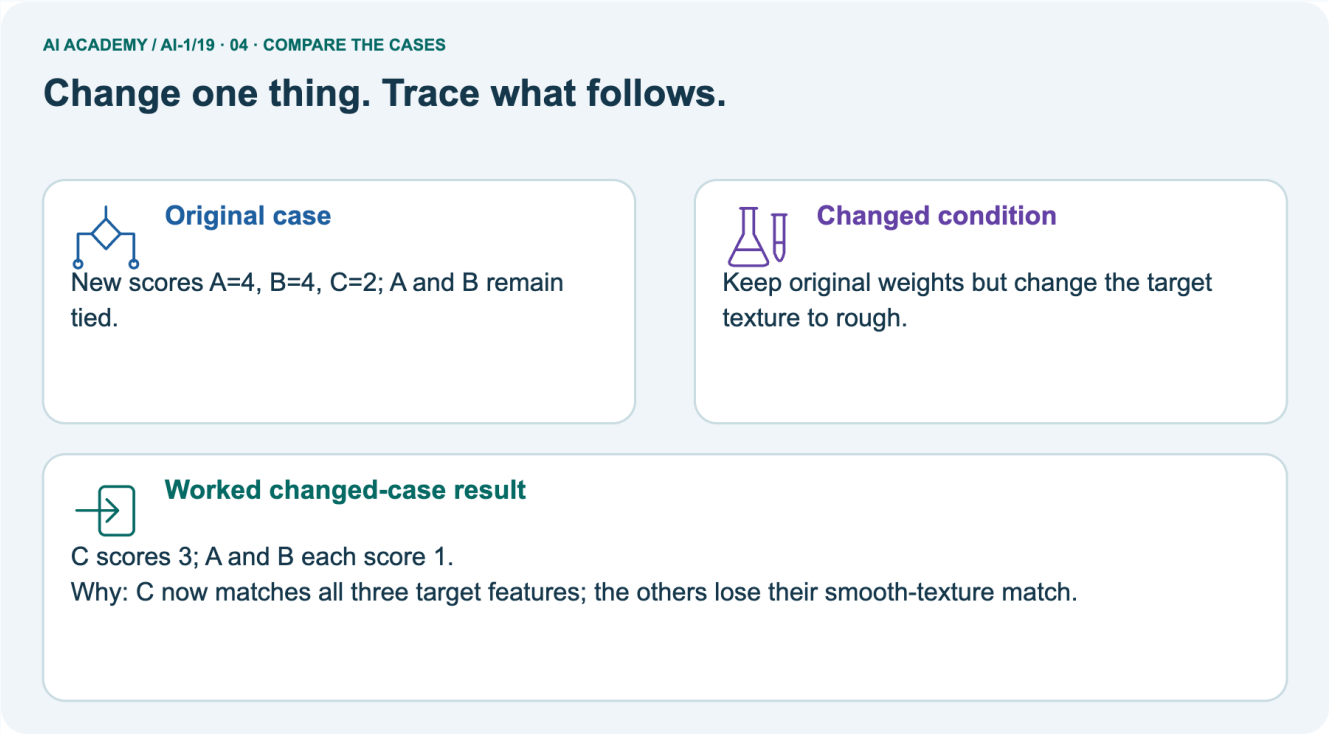
Follow the labelled picture with the matching explanation. Open the vector diagram

Declared **weight**s change **similarity**. A: All **weight**s 1 = 2, Texture **weight** 3 = 4. B: All **weight**s 1 = 2, Texture **weight** 3 = 4. C: All **weight**s 1 = 2, Texture **weight** 3 = 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

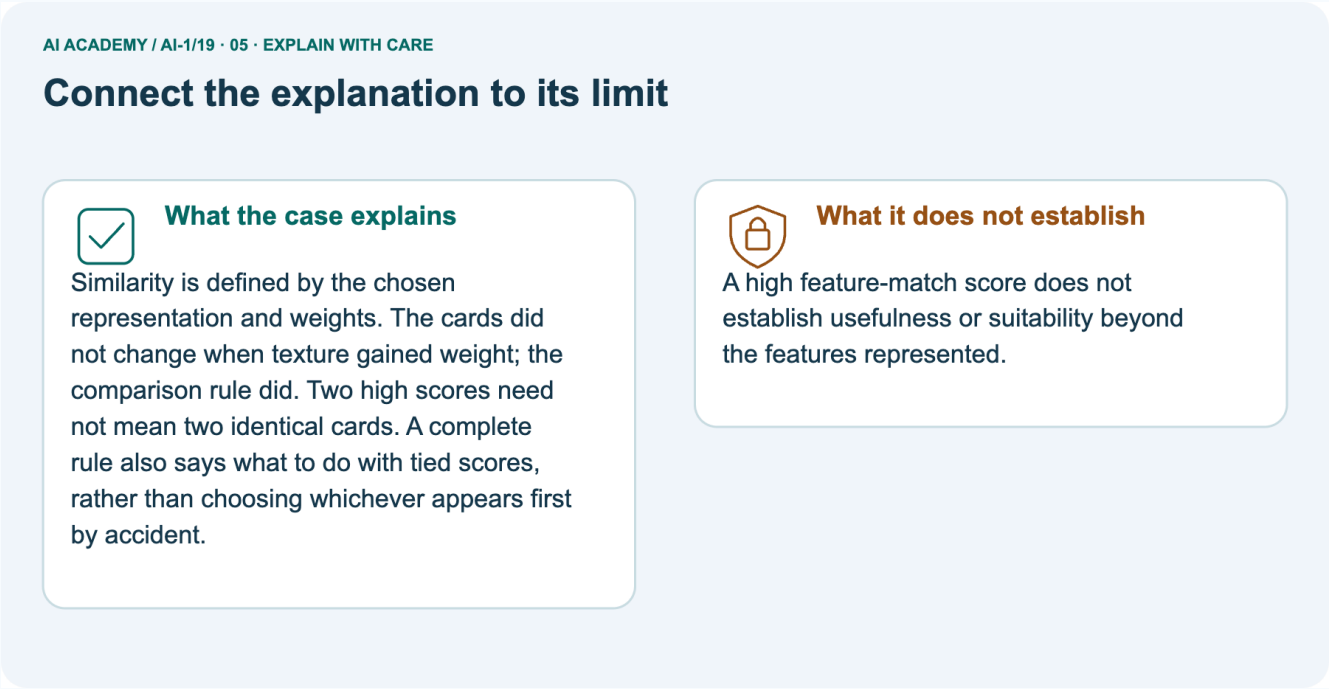
Original case: New scores A=4, B=4, C=2; A and B remain tied. Changed condition: Keep original **weight**s but change the target texture to rough. Expected result: C scores 3; A and B each score 1.

Reason: C now matches all three target features; the others lose their smooth-texture match.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

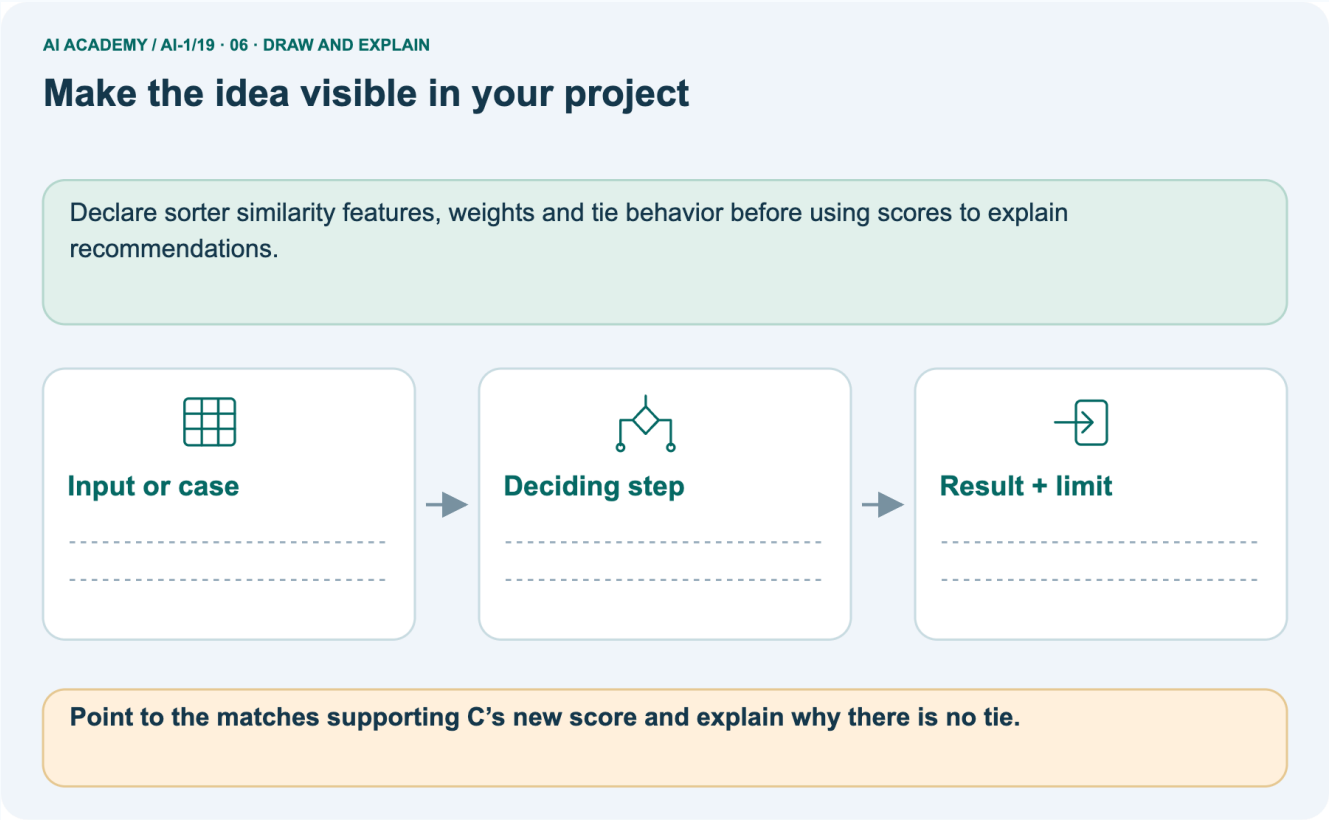
Similarity is defined by the chosen representation and **weight**s. The cards did not change when texture gained **weight**; the comparison rule did. Two high scores need not mean two identical cards.

A complete rule also says what to do with tied scores, rather than choosing whichever appears first by accident. Limit: A high feature-match score does not establish usefulness or suitability beyond the features represented.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Declare sorter **similarity** features, **weight**s and tie behavior before using scores to explain recommendations. Draw your **input**, deciding step, result and limit.

Supported question: Point to the matches supporting C's new score and explain why there is no tie.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.